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DESCRIPTION AND CHARACTERIZATION OF THE ALGAL SPECIES *AUREOUMBRA LAGUNENSIS* GEN. ET SP. NOV. AND REFERRAL OF *AUREOUMBRA* AND *AUREOCOCCUS* TO THE PELAGOPHYCEAE[1]

Hudson R. DeYoe[2] *and Dean A. Stockwell*

Marine Science Institute, 750 Channelview Drive, University of Texas at Austin, Port Aransas, Texas 78373

Robert R. Bidigare and Mikel Latasa

Department of Oceanography, 1000 Pope Road, University of Hawaii, Honolulu, Hawaii 96822

Paul W. Johnson

Electron Microscope Facility, University of Rhode Island, Kingston, Rhode Island 02881

Paul E. Hargraves

Graduate School of Oceanography, University of Rhode Island, Narragansett, Rhode Island 02882

and

Curtis A. Suttle

Department of Earth and Ocean Sciences, University of British Columbia, Vancouver, British Columbia, Canada V6T 1Z4

ABSTRACT

The Texas brown tide alga (strain TBA-2) is described as Aureoumbra lagunensis *Stockwell, DeYoe, Hargraves, et Johnson, gen. et sp. nov. Pigment composition, chloroplast structure, and 18S ribosomal RNA gene sequence data indicate that* A. lagunensis *and the east coast brown tide alga* Aureococcus anophagefferens *(originally placed in the Chrysophyceae) belong in the class Pelagophyceae. The new genus* Aureoumbra *with* A. lagunensis *as the type species differs from* Aureococcus *in 18S ribosomal RNA gene sequence, pyrenoid form, nitrogen physiology, and possession of basal bodies. The genus* Aureococcus *is placed in the order Pelagomonadales and family Pelagomonadaceae while ordinal placement of* Aureoumbra *is deferred.*

Key index words: Aureococcus anophagefferens; *basal bodies; east coast brown tide alga; 18S rRNA gene; harmful algal blooms; 19′-butanoyloxyfucoxanthin;* Pelagomonas calceolata, *Pelagophyceae,* Aureoumbra lagunensis *gen. et sp. nov., Texas brown tide alga*

The Texas brown tide alga and the east coast brown tide alga, *Aureococcus anophagefferens* Hargraves et Sieburth, are marine microalgae that have similar morphologies and produce nontoxic, but harmful algal blooms (Stockwell et al. 1993). *Aureococcus anophagefferens* was originally placed in the class Chrysophyceae based on its ultrastructure and pigmentation (Sieburth et al. 1988). Andersen et al. (1993), in describing the marine flagellate *Pelagomonas calceolata* Andersen et Saunders and the new class Pelagophyceae, suggested that *A. anophagefferens* may also belong in this class. Based on 18S ribosomal RNA gene sequence data, DeYoe et al. (1995) concluded that the Texas brown tide isolate, strain TBA-2, and *A. anophagefferens,* strain Pt-1, are both related to *P. calceolata,* but *A. anophagefferens* is more closely related to *P. calceolata.*

Here we formally describe the Texas brown tide alga (strain TBA-2) as *Aureoumbra lagunensis* Stockwell, DeYoe, Hargraves, et Johnson gen. et sp. nov., and suggest placement within the Pelagophyceae. Based on morphologic, structural, and sequence data, *Aureococcus anophagefferens* is also referred to the Pelagophyceae.

MATERIALS AND METHODS

Cultures. The strain of the Texas brown tide alga, TBA-2, used in this study was isolated by F. Chen in March 1991 from a bloom in Bird Island basin of Laguna Madre (27°28′ N, 97°18′ W) near Corpus Christi, Texas, using serial dilution (Guillard 1973) in f/2 medium (550 μM NO_3^-). Strain TBA-2 has been deposited in the Provasoli–Guillard National Center for Culture of Marine Phytoplankton as strain CCMP1681. *Aureococcus anophagefferens* Hargraves et Sieburth (Sieburth et al. 1988) Pt-1 culture was kindly provided by E. Cosper. Cultures were grown in unbuffered, sterile, artificial seawater (ESAW) (Harrison et al. 1980) modified as in DeYoe and Suttle (1994). Strain TBA-2 was grown at 25° C under constant illumination provided by Phillips cool-white fluorescent lamps (F96T12/CW/VHO) at 55 μmol quanta·m^{-2}·s^{-1}. *Aureococcus anophagefferens* was grown at 18°C at 68 μmol quanta·m^{-2}·s^{-1}. Growth irradiance (400–700 nm) was measured with a Li-Cor photom-

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[2] Present address and author for reprint requests: Biology Department, University of Texas–Pan American, Edinburg, Texas 78539-2999; e-mail hdeyoe@panam.edu.

eter (Lincoln, Nebraska) model LI-1000 equipped with a LI-193SA spherical quantum sensor.

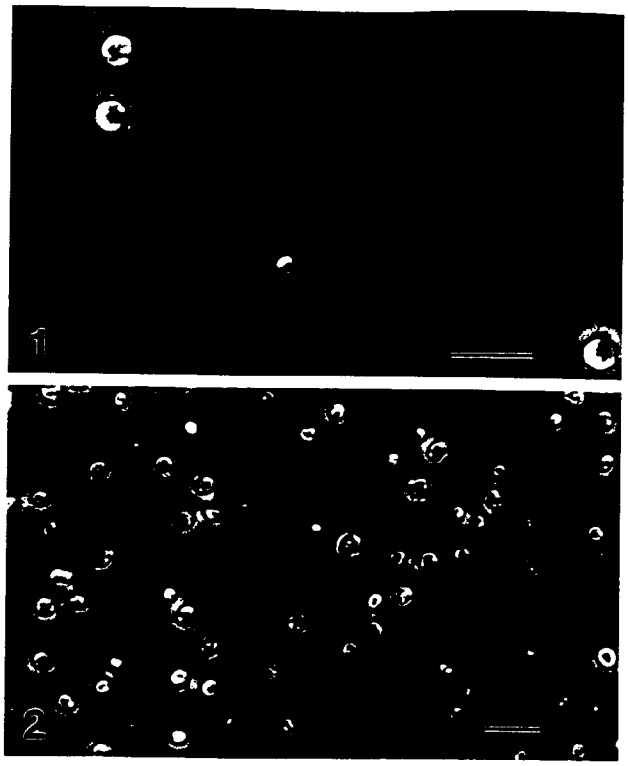
Field samples. Texas brown tide algae were collected for pigment analysis 24 May 1991 and 29 July 1994 in Laguna Madre, Texas, in areas of high bloom density (very low densities of other species were present). Samples were kept on ice until filtration in the laboratory under low light conditions. The field and culture samples were filtered at <10 mm Hg onto Whatman GF/F glass fiber filters (0.7 μm retention), wrapped in aluminum foil, and kept frozen. High-performance liquid chromatography (HPLC) pigment analyses were performed according to the procedure outlined in Wright et al. (1991).

Transmission electron microscopy. Cultured cells of strain TBA-2 (strains CCMP1508 and 1681) and cells from natural brown tide populations were fixed prior to concentration. Natural samples were fixed with glutaraldehyde (1% final concentration) and concentrated by settling and/or centrifugation. Approximately 10 mL of cultured cell concentrate were added to an equal volume of either 1% formaldehyde plus 3% glutaraldehyde in 0.1 M sodium cacodylate buffer (pH 7.2) with 3% NaCl, or 1% acrolein plus 2.5% glutaraldehyde in the same buffer and fixed at least 12 h at 4° C. One cultured cell sample fixed with the acrolein–glutaraldehyde fixative was stained with 1% ruthenium red. The fixed cells (natural or cultured) were centrifuged at 6000 × *g* for 30 min at 4° C. Cell pellets were gently separated into pieces and encased in molten 1.5% purified agar in buffer at 44° C, which was allowed to solidify at room temperature. Small agar cubes (1 mm^3) were cut around the pellet pieces and placed into 3% glutaraldehyde in 0.1 M sodium cacodylate buffer (pH 7.2) with 3% NaCl in 3-mL vials. After overnight fixation at 4° C, the samples were washed once with buffer with 3% NaCl, once with buffer with 1.5% NaCl, and once with buffer lacking NaCl. The samples were postfixed with 1% osmium tetroxide in buffer lacking NaCl for 3 h at 4° C and dehydrated in a graded ethanol series to absolute ethanol followed by two changes of propylene oxide. All samples were embedded in Spurr's low viscosity resin (Spurr 1969). Thin sections were cut with a diamond knife using a DuPont/Sorvall MT2-B ultramicrotome and stained with 2% uranyl acetate in 50% methanol followed by lead citrate (Venable and Coggeshall 1965). The sections were examined and photographed with a JEOL 1200EX transmission electron microscope operating at 80 kV accelerating voltage.

Light microscopy. Wet mounts of live cells in culture medium or seawater were prepared. Cells were selectively stained with alcian blue using the technique of Yokote et al. (1985).

RESULTS

Light and electron microscopy. Cells were solitary and spherical to subspherical with a golden color and ranged in diameter from 2.5 to 5.0 μm (Figs. 1, 2). In culture, cells were also oblong and occasional large cells occurred that were approximately twice the diameter of an average cell. Cultured cells sometimes had a small dimple-like depression. A representative cell is shown in Figures 1 and 3. One cup-shaped, parietal chloroplast with a single-stalked pyrenoid occupied most of the cell (Fig. 4). The thylakoid lamellae appeared to be three-layered and a girdle lamella was present (Figs. 4, 5). Evidence of a ring chloroplast genophore was lacking. The single Golgi body was large and often associated with a depression in the nuclear envelope (Fig. 6). The nucleus was typically ovoid, with one nucleolus evident (Fig. 6). The outer membrane of the nuclear envelope appeared confluent with the chloroplast. Flagella were not observed either emergent from or external to any cells (culture or natural samples). Two basal bodies, however, were commonly observed adjacent to a depression in the nuclear envelope (Fig. 7). These basal bodies were composed of a ring of nine microtubular triplets with no microtubules in the center.

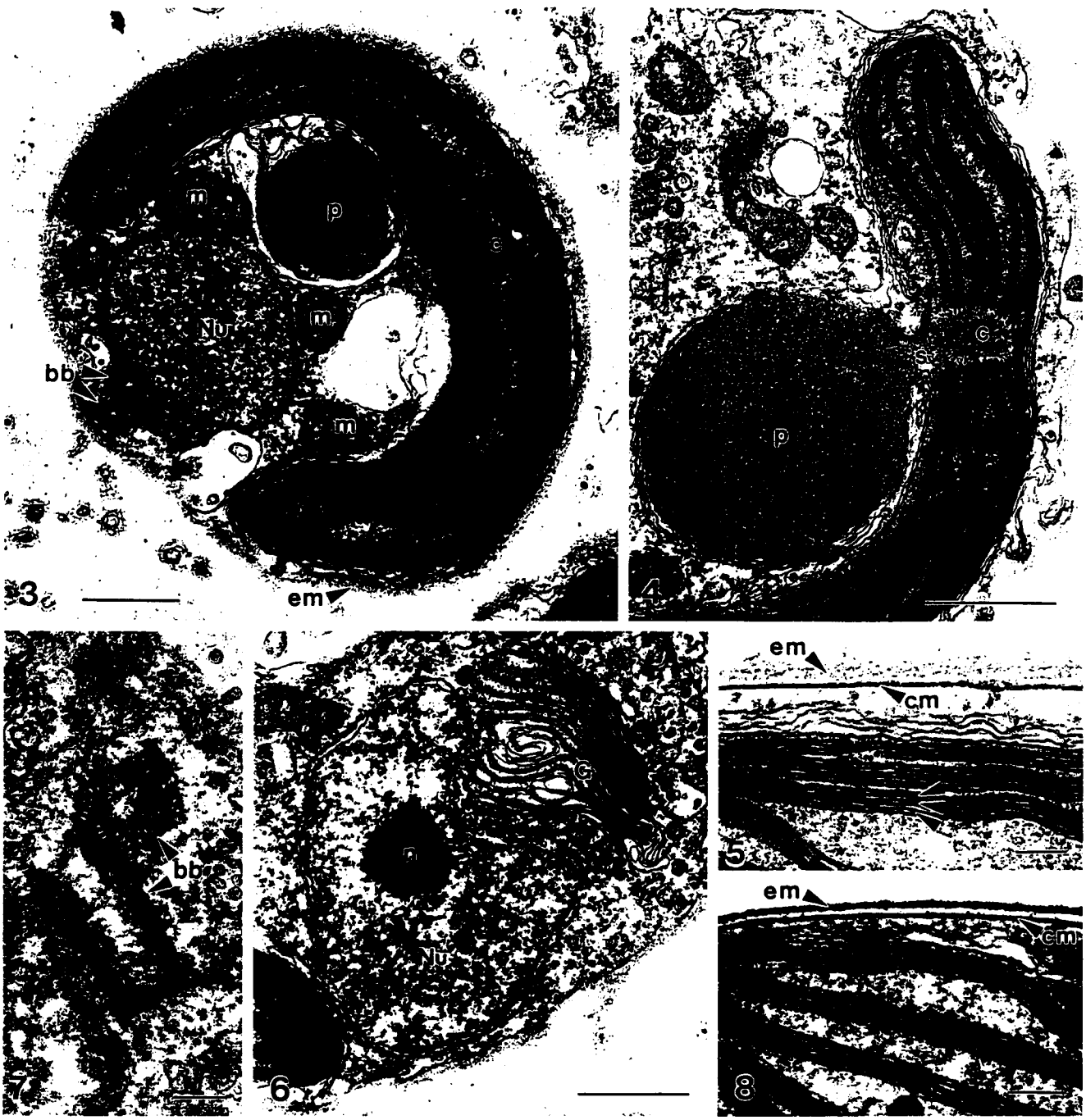


FIGS. 1–2. Morphology of *Aureoumbra lagunensis* as observed by phase contrast light microscopy. FIG. 1. Cells of culture strain TBA-2. FIG. 2. A fixed sample of cells from a bloom of the Texas brown tide, Laguna Madre, Texas, 14 November 1994. *A. lagunensis* cells dominate the sample. Scale bars = 10 μm.

Good fixation of cells both from natural populations and cultures proved to be difficult. Like *A. anophagefferens*, an organic matrix lies external to the plasmalemma (Fig. 5). Alcian Blue staining indicated that this extracellular matrix is most probably composed of polysaccharide. The extracellular matrix when stained with ruthenium red becomes a discrete electron-opaque layer (Fig. 8). In our samples, *A. lagunensis* lacked a cell wall.

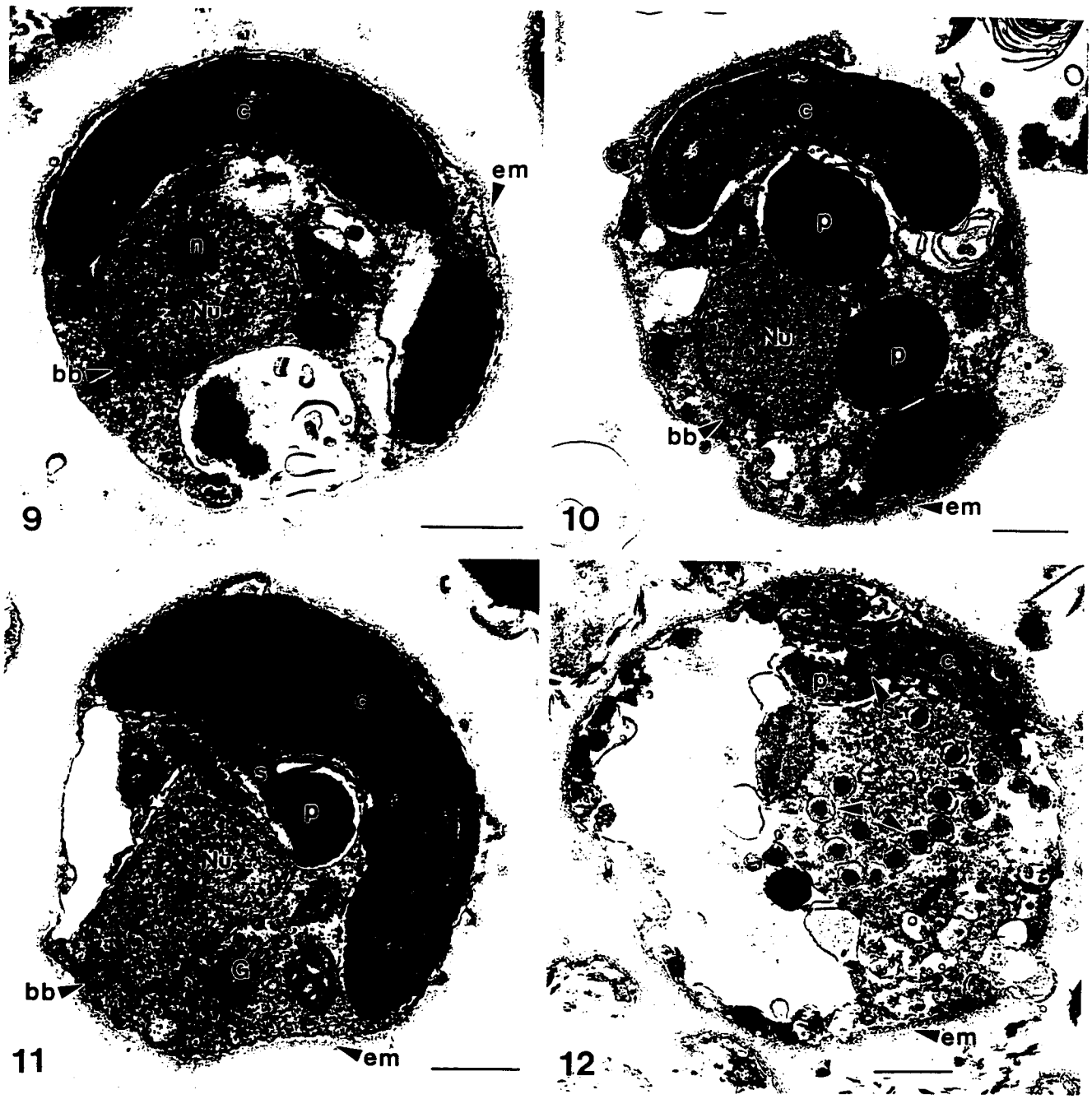
Aureoumbra lagunensis cells were observed by transmission electron microscopy (TEM) to be the dominant microalga on several occasions in samples from "brown tide" blooms in Upper Laguna Madre, Nueces Bay, and Copano Bay (Figs. 9–10) and had the same features seen in cultured cells (Fig. 11). Viral-like inclusions were observed in *A. lagunensis* cells in natural populations from Nueces Bay and Copano Bay in 1992 (Fig. 12).

Pigment data. The quantitatively important pigments identified in strains of *A. lagunensis* and *A. anophagefferens* were chlorophylls *a* and *c*, β,β-carotene, fucoxanthin, diatoxanthin, diadinoxanthin, and 19′-butan-



FIGS. 3–8. Ultrastructural characteristics of the type species *Aureoumbra lagunensis*, strain TBA-2, as observed by transmission electron microscopy of thin sections. FIG. 3. Holotype cell showing a single, cup-shaped chloroplast with an external pyrenoid, nucleus, mitochondria, pair of basal bodies, and extracellular material. FIG. 4. The external, stalked pyrenoid at its point of attachment to the chloroplast. The pyrenoid is surrounded by the girdle lamellae of the chloroplast. FIG. 5. The chloroplast membranes are three-layered (arrows). Cells show a loose layer of extracellular material external to the cytoplasmic membrane when fixed by conventional procedures. FIG. 6. The Golgi body is observed adjacent to a depression of the nucleus. FIG. 7. A pair of basal bodies oriented at approximately 90° to each other. FIG. 8. Cells fixed in the presence of 1% ruthenium red show condensed extracellular material. Scale bars in Figs. 3, 4, and 6 = 0.5 μm; Figs. 5, 7, and 8 = 0.1 μm. Abbreviations: bb, basal body; c, chloroplast; cm cytoplasmic membrane; G, Golgi body; em, extracellular material; Nu, nucleus; n, nucleolus; m, mitochrondion; p, pyrenoid; s, stalk.

oyloxyfucoxanthin (19′BF) (Table 1). The accessory pigment:chlorophyll *a* (Chl *a*) ratios were similar for both organisms with the exception of an order of magnitude lower 19′BF:Chl *a* ratio in *A. lagunensis* compared to *A. anophagefferens*. This difference between the algae was consistent even when comparing cultures sampled at different stages of growth (i.e. exponential vs. stationary phases).



FIGS. 9–12. Ultrastructure of *A. lagunensis* cells from natural populations (Figs. 9, 10, and 12) and in culture (Fig. 11), as observed by transmission electron microscopy of thin sections. FIG. 9. Cell from natural sample collected from Laguna Madre, March 1991. FIG. 10. Cell from natural sample from Laguna Madre, May 1994. FIG. 11. Cultured cell (strain CCMP 1508). FIG. 12. Cell containing viral-like inclusions (note electron-opaque cores) from Nueces Bay, October 1992. Scale bars = 0.5 μm. Abbreviations: bb, basal body; c, chloroplast; G, Golgi body; em, extracellular material; Nu, nucleus; n, nucleolus; p, pyrenoid; s, stalk; v, viral-like inclusion.

Culture and field samples of *A. lagunensis* taken during 1991 and 1994 displayed similar pigment compositions and accessory pigment: Chl *a* ratios (Table 1). Fucoxanthin : Chl *a* and diadinoxanthin : Chl *a* ratios of the 1991 field sample were slightly lower than those observed in culture and in the 1994 field sample.

DESCRIPTION

Aureoumbra Stockwell, DeYoe, Hargraves, et Johnson gen. nov.

Cellulae aureo-bruneae, sphaeroideae vel subsphaeroideae, uninucleatae, cum 2 granis basalibus, chloroplasto unico et pyrenoide pedicellato, sine stigmateque.

TABLE 1. *Accessory pigment : Chl a ratios (w/w) for* Aureoumbra lagunensis *(strain TBA-2 and natural populations),* Aureococcus anophagefferens *(strain Pt-1),* Pelagomonas calceolata *(strain MC-1 = CCMP 1214), and* Pelagococcus subviridis *(strain Pela C12 = CCMP 1429).*

Taxon	Accessory pigment : Chl *a* ratio[a]						Note[b]	Reference
	c:a	bf:a	f:a	dn:a	dt:a	car:a		
A. anophagefferens	0.42	0.21	0.54	0.05	nd[c]	0.01		b) Bidigare (1989)
A. anophagefferens	0.36	0.16	0.58	0.09	0.04	0.02	exp.	a) this study
A. anophagefferens	0.35	0.36	0.43	0.12	0.05	0.02	stat.	a) this study
A. lagunensis	0.13	0.02	0.24	0.12	0.01	0.02	exp.	a) this study
A. lagunensis	0.14	0.01	0.36	0.11	0.05	0.04	stat.	a) this study
A. lagunensis (natural population)	0.12	0.04	0.33	0.19	0.01	0.04	7/94	a) this study
A. lagunensis (natural population)	0.09	0.02	0.18	0.03	<0.01	0.02	5/91	a) this study
P. calceolata	0.22	0.29	0.49	0.06	nd	0.01		b) Bidigare (1989)
P. subviridus	0.32	0.74	0.22	0.10	nd	nd		b) Bidigare (1989)

[a] a, Chl *a*; c, total Chl- *c*-related pigments; bf, 19′-butanoyloxyfucoxanthin; f, fucoxanthin; dn, diadinoxanthin; dt, diatoxanthin; car, β,β carotene.
[b] exp., Sample harvested while cells in exponential growth; stat., sample harvested while cells in stationary phase; dates given are dates of sample collection in field.
[c] nd, Not detectable, below limit of quantification.

Cells golden brown, spherical or subspherical, uninucleate with two basal bodies, with a single chloroplast and stalked pyrenoid, lacking an eyespot. "Aureo" is derived from "aureus" meaning golden and "umbra" refers to the shading or reduced light penetration resulting from high concentrations of this alga in the Laguna Madre.

Aureoumbra lagunensis Stockwell, DeYoe, Hargraves, et Johnson sp. nov.

Characters generis exhibens; cellulae 2.5–5 μm diametro. Granum dictum "Golgi" in cytoplasto immersum; materia extracellularis polysaccharata adest. Cystae et stati quiescentes ignotae. Organulae expulsivae non visae. Locus typicus: Laguna Madre, Texas, U.S.A., 27°28′ Lat. Sept., 97°18′ Long. Occid. Holotypus = Figure 3.

With characters of the genus, cells 2.5–5.0 μm in diameter. Golgi body immersed in the cytoplasm. Extracellular polysaccharide-like material present. Cyst or resting stages unknown. Ejectile organelles not observed. The epithet "lagunensis" refers to the lagoon environment from which the alga was isolated. Type locality: Laguna Madre, Texas, U.S.A., 27°28′ N, 97°18′ W. Holotype = Figure 3.

Isotype material of *Aureoumbra lagunensis* (strain TBA-2) has been deposited with the University Herbarium, University of California, Berkeley. The type clone, strain TBA-2, has been deposited as strain CCMP1681 in the Provasoli–Guillard National Center for Culture of Marine Phytoplankton (CCMP) at the Bigelow Laboratory for Ocean Sciences (West Boothbay Harbor, Maine). Other isolates of *Aureoumbra lagunensis* are also available from the CCMP. The GenBank accession number for the 18S rRNA sequence of *Aureoumbra lagunensis* is U40258.

DISCUSSION

Aureoumbra lagunensis and *Aureococcus anophagefferens* are small planktonic marine algae that share a number of features including unicellularity, extracellular polysaccharide layer, chloroplast structure, and pigments. Based on available evidence, we propose that these algae be placed in the class Pelagophyceae with *Pelagomonas calceolata* and *Pelagococcus subviridis* (Saunders et al. 1995).

Aureoumbra lagunensis, A. anophagefferens, P. calceolata, and *P. subviridis* (Lewin et al. 1977) all possess Chl *a*, Chl *c*-related pigments, fucoxanthin, diatoxanthin, diadinoxanthin, and 19′BF, which are typical of Type III "aberrant chrysophytes" (Bidigare 1989) and members of the Pelagophyceae (Andersen et al. 1993). Type III "aberrant" chrysophytes differ from "true" (Type I) chrysophytes by the possession of 19′BF and the lack of the xanthophyll-cycle pigments, violaxanthin, antheraxanthin, and zeaxanthin (Withers et al. 1981, Bidigare 1989). A previous pigment analysis of *A. anophagefferens* (Bidigare 1989) tentatively identified the presence of diatoxanthin. The results of this study confirmed the presence of diatoxanthin, but at low levels in cultured and field-collected cells.

Similarity in pigment composition of the 1991 and 1994 field samples as well as the similarity of the 1994 field and culture samples suggests that the same species was sampled during 1991 and 1994, and that the type strain TBA-2 is the same as the bloom alga found in Laguna Madre in 1991 and 1994.

Aureoumbra lagunensis was the dominant species from Texas brown tide algal bloom samples taken on eight sampling dates over 4 years as determined by transmission electron microscopy. The distinctive ultrastructural features (chloroplast structure, stalked pyrenoid, and extracellular material) made it possible to recognize these cells in natural samples. The presence of virus-like particles in *A. lagunensis* cells in samples from Nueces and Copano Bays (Fig. 12) suggests the potential for viral control of this bloom-forming alga. Viral control has been suggested for the east coast brown tide alga *Aureo-*

coccus anophagefferens (Sieburth et al. 1988, Nixon et al. 1994).

Aureococcus anophagefferens and *Aureoumbra lagunensis* possess the features of the class Pelagophyceae (Saunders et al. 1997), which include chloroplast lamellae with three adpressed thylakoids and a girdle lamella, the presence of fucoxanthin, diatoxanthin, diadinoxanthin, and 19'BF, and lack of both an eyespot and violaxanthin. Based on our electron micrographs, evidence of a ring-shaped DNA genophore in the chloroplast of *A. lagunensis* is lacking, which is consistent with the emended description of the Pelagophyceae of Saunders et al. (1997). Sequence data of the 18S ribosomal RNA gene strongly support the incorporation of *A. anophagefferens* and *A. lagunensis* into the Pelagophyceae (DeYoe et al. 1995). This class includes *Pelagococcus subviridis* (Saunders et al. 1995) and the flagellate *Pelagomonas calceolata* (Andersen et al. 1993). Ultrastructural, pigment, and molecular information support including *A. lagunensis* in the Pelagophyceae and reassigning *A. anophagefferens* from the Chrysophyceae to the Pelagophyceae.

Although similar in appearance, *A. lagunensis* and *A. anophagefferens* differ in cell size (Stockwell et al. 1993), pyrenoid form, pigment ratios, nitrogen physiology (DeYoe and Suttle 1994), growth temperature optima (DeYoe, unpubl.), and most notably, in 18S rRNA gene sequences (DeYoe et al. 1995). Furthermore, *A. lagunensis* possesses a pair of basal bodies that is lacking in *A. anophagefferens*. Based on these data, it is clear that *A. lagunensis* and *A. anophagefferens* are not the same organism, and the erection of the genus *Aureoumbra* is justified.

One unresolved feature of *A. lagunensis* is the presence or absence of a cell wall. In our micrographs of cells from natural and cultured samples, a cell wall was lacking; however, other investigators have observed a wall in micrographs of cells from other natural populations (Robert Andersen, pers. comm.). It is possible that environmental and physiologic fluctuations may affect the deposition of organic material on the external cell surface, leading to the appearance of a cell wall only under certain but as yet undefined conditions.

Saunders et al. (1997) have recently emended the description of the class Pelagophyceae as well as the associated order and family and have tentatively referred the order Sarcinochrysidales to the Pelagophyceae based on their 18S rRNA sequence analysis. In this analysis, *Aureococcus* falls in the same clade with *Pelagomonas caleolata*. As *A. anophagefferens* has the features of the Pelagomonadales and Pelagomondaceae sensu Saunders et al. (1997), placement of *Aureococcus* in this order and family is suggested. In the same analysis, *A. lagunensis* along with the isolate CCMP 1410 fall into the same clade with two representatives of the Sarcinochrysidales (*Sarcinochrysis* and *Pulvinaria*) (Saunders et al. 1997). *Aureoumbra lagunensis*, except in one unconfirmed report (Robert Andersen, pers. comm.), lacks flagella even though it has basal bodies. Lacking flagella, *Aureoumbra* cannot be placed in the Sarcinochrysidales sensu stricto (Gayral and Billard 1977). Inclusion of this genus in the Pelagomonadales would be contrary to the sequence-based phylogeny of Saunders et al. (1997), although the current description of the Pelagomonadales would allow it. To erect a new order based on inadequate knowledge of one described taxon, *Aureoumbra*, seems unwise; therefore, we suggest delaying ordinal placement of this taxon until more morphologic and ultrastructural data are obtained and other potential sister taxa are described.

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Andersen, R. A., Saunders, G. W., Paskind, M. P. & Sexton, J. P. 1993. Ultrastructure and 18S rRNA gene sequence for *Pelagomonas calceolata* gen. et sp. nov. and the description of a new algal class, the Pelagophyceae *classis nov. J. Phycol.* 29: 701–15.

Bidigare, R. R. 1989. Photosynthetic pigment composition of the brown-tide alga: unique chlorophyll and carotenoid derivatives. *In* Cosper, E. M., Bricelj, V. M. & Carpenter, E. J. [Eds.] *Novel Phytoplankton Blooms: Causes and Impacts of Recurrent Brown-tides and Other Unusual Blooms.* Springer-Verlag, New York, pp. 57–75.

DeYoe, H. R., Chan, A. M. & Suttle, C. A. 1995. Phylogeny of *Aureococcus anophagefferens* and a morphologically similar bloom-forming alga from Texas as determined by 18S ribosomal RNA sequence analysis. *J. Phycol.* 31:413–8.

DeYoe, H. R. & Suttle, C. A. 1994. The inability of the Texas "brown-tide" alga to use nitrate-nitrogen and the role of nitrogen in the initiation of a persistent bloom of this organism. *J. Phycol.* 30:800–6.

Gayral, P. & Billard, C. 1977. Synopsis du nouvel ordre des Sarcinochysidales (Chrysophyceae). *Taxon* 26:241–5.

Guillard, R. R. L. 1973. Isolation and purification: methods for microflagellates and nannoplankton. *In* Stein, J. R. [Ed.] *Handbook of Phycological Methods: Growth and Culture Measurements.* Cambridge University Press, Cambridge, pp. 69–86.

Harrison, P. J., Waters, R. E. & Taylor, F. J. R. 1980. A broad spectrum artificial seawater medium for coastal and open ocean phytoplankton. *J. Phycol.* 16:28–35.

Lewin, J., Norris, R. E., Jeffrey, S. W. & Pearson, B. E. 1977. An aberrant chrysophycean alga *Pelagococcus subviridus* gen. nov. et sp. nov. from the north Pacific Ocean. *J. Phycol.* 13:259–66.

Nixon, S. W., Granger, S. L., Taylor, D. I., Johnson, P. W. & Buckley, B. A. 1994. Subtidal volume fluxes, nutrient inputs and the brown tide—an alternate hypothesis. *Estuarine, Coastal and Shelf Sciences* 39:303–12.

Saunders, G. W., Potter, D. & Andersen, R. A. 1997. Phylogenetic affinities of the Sarcinochrysidales and Chrysomeridales (Heterkonta) based on analyses of molecular and combined data. *J. Phycol.* 33:310–8.

Saunders, G. W., Potter, D., Paskind, M. P. & Andersen, R. A. 1995. Cladistic analyses of combined traditional and molecular data sets reveal an algal lineage. *Proc. Natl. Acad. Sci. USA* 92:244–8.

Sieburth, J. M., Johnson, P. W. & Hargraves, P. E. 1988. Ultrastructure and ecology of *Aureococcus anophagefferens* gen. et sp. nov. (Chrysophyceae): The dominant picoplankter during a bloom in Narragansett Bay, Rhode Island, Summer 1985. *J Phycol.* 24:416–25.

Spurr, A. R. 1969. A low-viscosity epoxy resin embedding medium for electron microscopy. *J. Ultrastruc. Res.* 26:31–43.

Stockwell, D. A., Buskey, E. J. & Whitledge, T. E. 1993. Studies on conditions conducive to the development and maintenance of a persistent "brown-tide" in Laguna Madre, Texas. *In* Smayda, T. J. & Shimizu, Y. (Eds.) *Toxic Phytoplankton Blooms in the Sea.* Elsevier Science Publishers, Amsterdam, pp. 693–8.

Venable, J. H. & Coggeshall, R. 1965. A simplified lead citrate stain for use in electron microscopy. *J Cell Biol.* 25:407–8.

Withers, N., Fiksdahl, A., Tuttle, R. C. & Liaaen-Jensen, S. 1981. Carotenoids of the Chrysophyceae. *Comp. Biochem. Physiol.* 68B:345–9.

Wright, S. W., Jeffrey, S. W., Mantoura, R. F. C., Llewellyn, C. A., Bjornland, T., Repeta, D. & Welschmeyer, N. 1991. Improved HPLC method for the analysis of chlorophylls and carotenoids from marine phytoplankton. *Mar. Ecol. Prog. Ser.* 77:183–96.

Yokote, M., Honjo, T. & Asakawa, M. 1985. Histochemical demonstration of a glycocalyx on the cell surface of *Heterosigma akashiwo. Mar. Biol.* 88:295–9.